

The Milky Way's Hidden Mass: Measuring Galactic Rotation with 129 Million Stars

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The question: planets far from the Sun orbit slower than near ones. Do stars far from the Galaxy's centre do the same? If they don't, there must be unseen mass holding them, that is dark matter. We test this with Gaia DR3, which is far too big for Excel or pandas, so we profile it on the command line, break the in-memory tools, then compute the answer with PySpark on a cluster.

Profiling

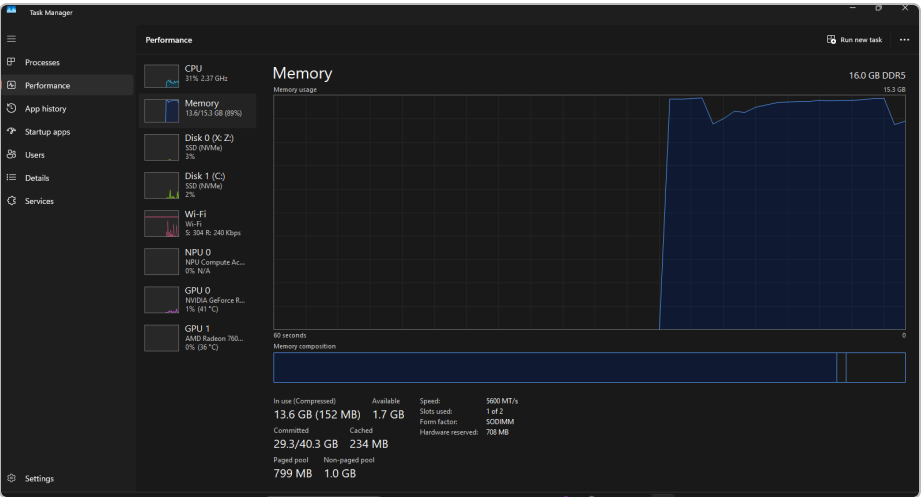
Our slice: 242 of 3,386 files (every 28th, all-sky), 53.7 GB, **129,016,649 rows**, 152 columns.

Command (streamed, timed)	Result
du -sh	26 GB (121-file draft)
zcat grep -v'^#' wc -l	64.7M rows (draft)
awk mean parallax	0.943 mas
grep -c VARIABLE	5,345

Memory stayed at a **few MB** no matter the file size. These tools stream, they never load.

Breaking

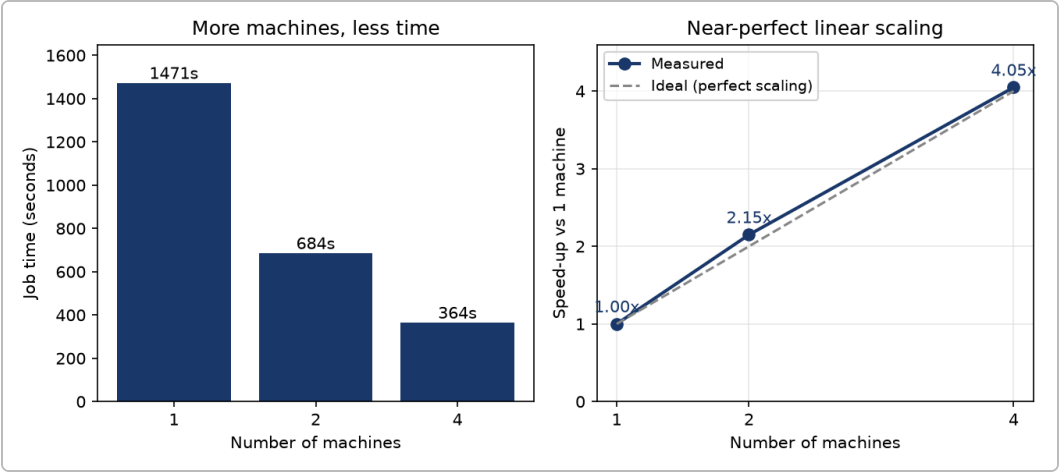
Excel: hangs opening one raw file; hard cap 1,048,576 rows. Full data is **~120x** that.
pandas: loading the data drove committed memory to **29.3 GB** on a 15.3 GB laptop:



Mean parallax	Result
CLI awk (streams)	0.943, done
pandas (loads)	out of memory

Scaling

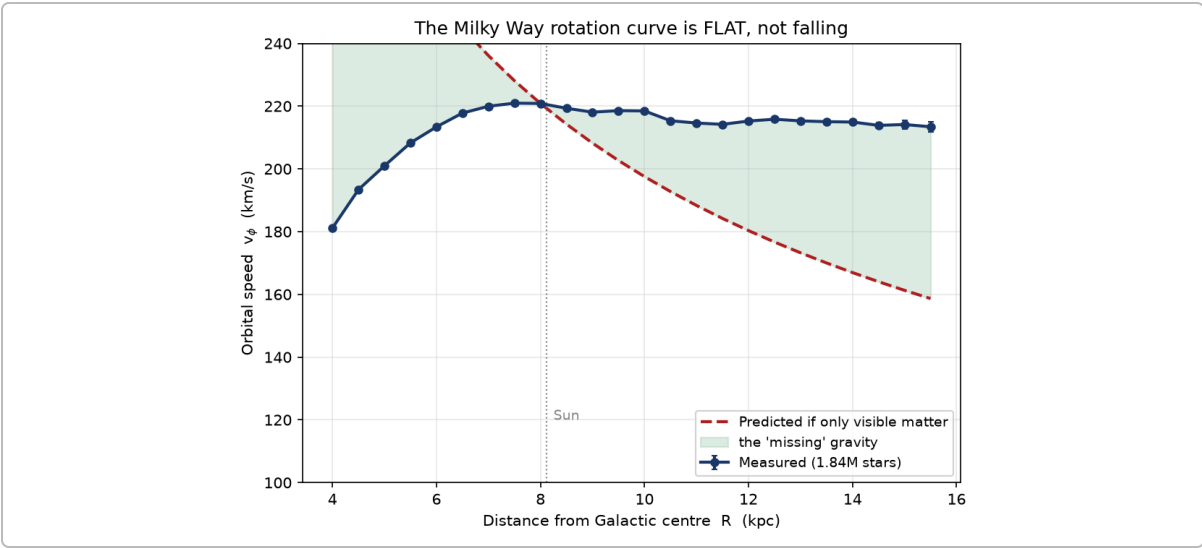
Same PySpark job on Dataproc, reading 53.7 GB straight from Google Cloud Storage. Only the machine count changes.



Machines	Time	Speed-up
1	1471 s	1.00x
2	685 s	2.15x
4	364 s	4.05x

Near-linear: 242 files = 242 tasks, always more than the cores, so every machine stayed busy.

The Answer: the Milky Way is full of Dark Matter



The result: from 1.84M quality stars, the orbital speed stays **flat at ~215 km/s out to 15 kpc**. Visible matter alone predicts the red curve, a fall to ~159 km/s. The green gap is gravity from mass we cannot see, that is **dark matter**. Roughly two-thirds of the mass inside 15 kpc emits no light.

Validation: with no tuning, the pipeline returns $v_\phi = 220.9$ km/s at the Sun's radius, matching the known value (astropy agrees to 0.06 km/s); inward and vertical motions average zero.

Scope: the parallax quality cut keeps nearby stars, so we reach ~15 kpc. The flat trend over that range already rejects the visible-matter-only prediction.

